

An evaluation model for indoor environmental quality (IEQ) acceptance in residential buildings

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ABSTRACT

The indoor environmental quality (IEQ) in residential buildings is examined from the prospect of an occupant's acceptance in four aspects: thermal comfort, indoor air quality, noise level and illumination level. Based on the evaluations by 125 occupants living in 32 typical residential apartments in Hong Kong, this study proposes empirical expressions to approximate the overall IEQ acceptance with respect to four contributors, namely operative temperature, carbon dioxide concentration, equivalent noise level and illumination level, via a multivariate logistic regression model. A range of IEQ acceptances for regular residential conditions is determined and the dependence of the predicted overall IEQ acceptance on the variations of the contributors is discussed. The proposed overall IEQ acceptance can be used as a quantitative assessment criterion for similar residential environments where an occupant's evaluation is expected.

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1. Introduction

Indoor environmental quality (IEQ) and occupant comfort are closely related. Current indoor environmental assessment includes four aspects, namely thermal comfort (TC), indoor air quality (IAQ), visual comfort (VC) and aural comfort (AC) [1–2].

Maintaining satisfactory thermal comfort conditions for the occupants by an adjustable indoor temperature set point of the air-conditioning system is one of the primary concerns in many air-conditioned office buildings. Thermal comfort relates human sensation and perception with a number of environmental and physical parameters [3]. It is by definition the perception of satisfaction a subject experiences in a given thermal environment [4]. Extensive studies resulted in a number of thermal comfort equations as proposed in some widely used design guides and standards [4–5]. Three indices were derived based on Fanger's comfort equation: Predicted Mean Vote (PMV), Predicted Percentage Dissatisfied (PPD) and Lowest Possible Percentage Dissatisfied (LPPD). The PMV and PPD indices are comparatively common in practical applications [6–7]. The former predicts the mean thermal comfort votes among a large group of people; the latter is a quantitative measure of the number of thermally dissatisfied persons in a group under particular thermal conditions.

Field studies on the thermal comfort of occupants working in an air-conditioned environment can be used to examine the neutral

temperature—a temperature associated with a neutral thermal sensation [8–10]. This temperature is a key factor for selecting an appropriate air temperature set point for an indoor thermal environment [11]. As climate, besides occupant factors including lifestyle, economic status and adaptive behaviour, plays an important role in affecting the indoor thermal environment [12–13], neutral temperatures for different climatic zones have been studied [1,10,14].

IAQ, as the nature of air in an indoor environment with relation to occupant health and comfort, is neither a simple nor an easily defined concept. In a broad context, it is the result of the complex interactions among buildings, building systems and people. Comparative risk studies performed by the United States Environmental Protection Agency (USEPA) ranked IAQ one of the top five environmental risks to public health [15]. Over the past decades, exposure to indoor air pollutants is believed to have increased due to a variety of factors, including the construction of more tightly sealed buildings, the reduction of ventilation rates (for energy saving), and the use of synthetic building materials and furnishings as well as chemically formulated personal care products, pesticides and household cleaners. As investigating all types of indoor air pollutants for general air quality monitoring and assessment is a complicated matter [16–18], it was suggested that the measurement and analysis of indoor carbon dioxide (CO₂) concentration could be useful for understanding IAQ and ventilation effectiveness [19–21]. Although healthy people can tolerate a CO₂ level up to 10,000 ppm without serious health effects, an acceptable indoor CO₂ level should be kept below 1000 ppm or 650 ppm above the ambient level in order to prevent any

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Nomenclature

$C_{0,0}$ and $C_{i,0}$	regression constants
C_L	clothing values (clo)
C_{op}	constant for the air velocity
GM	geometric mean
GSD	geometric standard deviation
M_e	metabolic rates (met)
n_{ij}	non-zero counts
N	survey sample size
$N_{\phi_0=1}$	acceptance count
R	coefficient
R_h	relative humidity (%)
SD	standard deviation
T_a	indoor air temperature (°C)
T_{op}	operative temperature (°C)
T_r	radiant temperature (°C)
V_a	air velocity (ms^{-1})

Greek letters

ϕ_0	overall IEQ acceptance
ϕ_i	occupant acceptance
ϕ_1	thermal environment acceptance
γ	PMV index
ζ_1	thermal sensation vote
ζ_2	CO ₂ concentration (ppm)
ζ_3	horizontal illumination level (lx)
ζ_4	sound pressure level (dBA)

accumulation of associated human body odour [22–23]. In terms of occupant satisfaction, acceptable IAQ means room air in which no known contaminants are at harmful concentration levels and at least 80% of the people exposed to it do not express any dissatisfaction [22–24].

Light enables humans to see. A number of available guides and codes of practice provide recommendations on adequate indoor lighting designs. For example, an illumination level of 2000 lx with a colour rendering index not less than 90 is required for a fabric inspection factory while 500 lx with a colour rendering index between 60 and 80 should be maintained in a general office [25]. Lighting quality attributed to the quantity and colour spectrum of light can be expressed by a comprehensive comfort, satisfaction, and performance (CSP) index [26]. Since indoor visual comfort is closely related to the horizontal and cylindrical illumination levels, adjustment of the illumination level is essential to improve visual comfort and occupant acceptance [27].

All sounds that are distracting, annoying, or harmful to everyday activities such as work, rest, study and entertainment are regarded as noises. In fact, any sound judged undesirable by the recipient can be considered a noise. Noise can be continuous or impulsive and both types can cause adverse effects on physical, mental and social well-being. A number of measures were proposed for indoor aural comfort evaluation, for instance, the equivalent sound pressure level (SPL), the noise criterion (NC) curves, the balanced noise criterion (BNC), the noise rating (NR), the preferred noise criterion (PNC), the room criterion (RC), and the loudness level. Given the nature of noises generally encountered in offices, a previous research indicated the A-weighting equivalent sound pressure level that has been widely adopted in the studies of noise level within buildings would be the best and most convenient measure, although the loudness level would also be a very good alternative [28]. In Hong Kong, surveys showed that

the equivalent continuous noise level correlated well with the occupant acceptance reported in air-conditioned offices [29], and in construction site offices where the background noise was not dominated by air-conditioning but by outdoor noise sources [30]. The neutral sound pressure level found for aural comfort in some typical air-conditioned offices was between 45 and 70 dBA, with a mean of 57.5 dBA.

Studies showed that an occupant's acceptance of an acceptable IEQ environment would be closely related to aforesaid environmental parameters. More recently, the equivalence of the discomfort caused by different physical qualities for indoor air quality, thermal sensation and noise has been investigated [31–34]. It was reported that every 1 °C change of operative temperature in range between 23 and 29 °C would produce the same feeling on human comfort due to a deviation 2.4 decipol regarding the perceived air quality [31], or due to a change in noise level of 3.9 dB. For levels of perceived air quality up to 10 decipol, a 1 decipol change in perceived air quality had the same effect on human comfort as a change in noise level of 1.2 dB [1,32]. The equivalence between acoustic and thermal sensation was proposed for short-term exposure as a 1 °C change in temperature had the same effect on 2.6 dBA [33]. Workplace variables inducing the largest number of health symptoms, comfort or odour concerns were investigated by multivariate regression analysis [34–38]. It was realised that successful control of the indoor environment required an understanding of the integral indoor environmental parameters. The occupants' acceptance of the four basic IEQ components was evaluated and correlated with the overall IEQ acceptance of an office environment [2]. Mathematical expressions were proposed for the overall IEQ acceptance using a multivariate logistic regression model and it can be used as a quantitative measure for an office environment design.

IEQ parameters are interdependent and must be considered interactively. However, conflicts were reported in above cases where these parameters were treated discretely, e.g. maximization of openings for natural ventilation and daylight resulted in poor acoustic performance and thermal discomfort in an indoor environment in subtropical climates [39]. Laboratory experiments of controlled auditory and visual stimuli also showed that the visual parameter was predominant in audio–visual interactions and the visual information would affect the auditory judgment [40]. Reportedly, attention to a visual form would reduce the conscious perception of sound, and vice versa when the sound was related to a scene [41]. In fact, interactions between visual and auditory perception give people a sense of involvement.

This study investigated the occupant acceptance of residential IEQ through physical measurements and subjective surveys. Mathematical expressions were proposed for the overall IEQ acceptance using a multivariate logistic regression model with the four environmental parameters discussed above. The study provides useful information for developing quantitative assessments for residential environments where an occupant's evaluation is expected.

2. Indoor environmental quality

Latent occupant responses of acceptance for individual environmental conditions were reported in previous studies on thermal comfort [10], IAQ [23], visual environment [27] and aural environment [29] in workplaces. Followed with the assumptions of the binary latent response that occupant preferred some environmental conditions in an environment, this study extends the concept of the overall IEQ acceptance to residential buildings. However, the correlations of the occupant preference would be invalid for residential buildings for as the variations of environmental parameters in the office and residential environment were

different. Regression coefficients can be evaluated with surveyed occupant responses from residential environment for the overall IEQ acceptance. The overall IEQ acceptance ϕ_0 can be expressed by a multivariate logistic regression model as shown in Eq. (1) [2], where $C_{0,0}$ and $C_{i,0}$ are the regression constants which can be determined from field measurements, ϕ_i is the occupant acceptance correlated with the thermal sensation vote ζ_1 , the CO₂ concentration ζ_2 (ppm), the horizontal illumination level ζ_3 (lx) and the equivalent sound pressure level ζ_4 (dBA).

$$\phi_0 = 1 - \frac{1}{1 + \exp(C_{0,0} + \sum_{i=1}^4 C_{i,0} \phi_i(\zeta_i))} \quad (1)$$

The thermal environment acceptance $\phi_1(\zeta_1)$, with the maximum acceptance = 0.95, is given below [3], where $C_{0,1}$ and $C_{1,1}$ are the regression coefficients,

$$\phi_1 = 0.95 \exp[-(C_{0,1} \zeta_1^2 + C_{1,1} \zeta_1^4)] \quad (2)$$

The acceptances $\phi_2(\zeta_2)$, $\phi_3(\zeta_3)$ and $\phi_4(\zeta_4)$ are expressed by logistic regression models with regression coefficients $C_{0,j}$ and $C_{1,j}$,

$$\phi_j = 1 - \frac{1}{1 + \exp(C_{0,j} + C_{1,j} \zeta_j)}; \quad j = 2, \dots, 4 \quad (3)$$

3. Field measurements

Subjective as well as objective evaluations of indoor environmental conditions made by 125 occupants (82 females and 43 males) from 32 typical residential apartments in Hong Kong were collected through individual interviews. The interviewees were mainly those occupants staying at home the longest time (as compared with other activities), and the housing samples covered various dwelling classes: 1 luxury, 13 private, 14 public; the remaining 4 were Home Ownership Scheme (HOS) flats. It was noted that the private housing flats were built for selling purposes but the public ones were government funded low-cost housing rented to families of lower income. The inclusion of various apartment types could cover a wide range of probable indoor environmental conditions. The apartments varied in size from 33 to 102 m² and were equipped with window or split air-conditioners. The only exception was the 102 m² dwelling which was centrally air-conditioned.

The indoor physical parameters describing the indoor environmental quality of a space were measured by seven separate meters for indoor air temperature (°C), globe temperature (°C), air velocity (m s⁻¹), relative humidity (%), CO₂ concentration (ppm), horizontal illuminance level (lx) and sound pressure level (dBA). Each measurement was carried out for fifteen minutes and the logged data were averaged over this designated time interval. Unlike the

IEQ measurements made for air-conditioned offices at a level of occupant's working plane, the sampling location in this study was set at the activity area of each interviewee. Therefore, the measured IEQ data could sufficiently reflect the real-time exposed indoor environment to the respondent. The number of measurements was determined based on the arrangement and partitioning of the specific apartment and the distribution of occupants. In an example case of totally four residents whom three of them were watching television in a combined dinning and living room (dinning/living room) and one of them was reading book in bedroom. Unlike the measurement approach in office, each assessment sample in residential housing did not necessarily require a separate 15 min physical measurement due to the small living area in Hong Kong. In this case, two physical measurements carried out in both dinning/living room and bedroom were considered representative for the assessment of the example apartment case. The IEQ data logged in the dinning/living room were regarded to be applicable to the individual occupant within the group. Such condition was judged depending on the separation distance between each occupant. Being the base for evaluating the energy benchmarking models, effective measurements were essential and thus accurate and reliable data could be obtained.

Via a dichotomous assessment scale, the occupant acceptance of the perceived indoor environment was recorded in the form of direct feedback using the question 'Is the thermal environment/indoor air quality/noise level/illumination level being perceived in the residential environment acceptable to you?' [42] The ranks '(1) Yes, acceptable' and '(0) No, not acceptable' were self-explanatory. In order to confirm the validity of a response, each respondent had to use a semantic differential evaluation scale for the subjective assessment of thermal environment and IAQ, and a visual analogue assessment scale for the evaluation of aural and visual comforts. At the end of the survey, an overall IEQ acceptance was determined.

4. Results and discussions

The interviewees were in the age range of 10–60, with an average age of 32 (standard deviation SD = 12). Their respective clothing values C_l (clo), metabolic rates M_e (met), dwelling activities and personal judgments about the environmental quality were registered. Taking 1 met = 58.2 W m⁻² for an average person of body surface area of 1.86 m² and 1 clo = 0.155 m² °C W⁻¹, the statistical values of M_e and C_l are as shown in Table 1. The table also exhibits the local environmental conditions described by the following parameters: indoor air temperature T_a (°C), radiant temperature T_r (°C), relative humidity R_h (%), air velocity V_a (m s⁻¹), CO₂ concentration ζ_2 (ppm), horizontal illumination level ζ_3 (lx), equivalent noise level ζ_4 (dBA), and operative temperature T_{op} (°C) which can be calculated by Eq. (4), where C_{op} is a constant for the

Table 1
Indoor environmental quality in surveyed residential buildings.

	Parameter									
	T_a (°C)	T_r (°C)	R_h (%)	V_a (m s ⁻¹)	T_{op} (°C)	ζ_2 (ppm)	ζ_3 (lx)	ζ_4 (dBA)	M_e (met)	C_l (clo)
'Acceptable' IEQ (N = 116)										
Average	27.3	27.4	84	0.36	27.4	689	178	67.1	1.05	0.48
SD	2.2	1.9	10.5	0.19	2.0	328	253	6.0	0.11	0.11
Minimum	23.0	22.5	67	0.08	22.8	326	4	52.0	1.0	0.10
Maximum	31.5	31.3	100	0.75	31.4	1499	1522	77.9	1.6	0.67
'Unacceptable' IEQ (N = 9)										
Average	28.1	28.1	84	0.49	28.1	497	307	70.6	1.11	0.49
SD	2.5	2.6	10.9	0.27	2.5	293	461	8.4	0.14	0.12
Minimum	23.0	22.5	67	0.09	22.8	326	33	52.0	1.0	0.29
Maximum	31.4	31.3	97	0.75	31.4	1045	1522	77.9	1.3	0.57
p-value (t-test)	0.32	0.45	0.96	0.15	0.37	0.06	0.41	0.22	0.22	0.70

air velocity [4],

$$T_{op} = C_{op}T_a + (1 - C_{op})T_r; \quad C_{op} = \begin{cases} 0.5; & V_a < 0.2 \\ 0.6; & 0.2 \leq V_a < 0.6 \\ 0.7; & 0.6 \leq V_a \end{cases} \quad (4)$$

Indoor environments in residential buildings are generally controlled to satisfy the occupants. Hence, it is not surprising that most interviewees (116 out of 125) were satisfied with the overall IEQ. For individual environmental parameters, except CO₂ concentration ($p \leq 0.06$, t -test), no significant differences were reported between acceptable and unacceptable IEQs ($p \geq 0.05$, t -test).

4.1. Thermal comfort

The dichotomous assessment showed that a total of 112 interviewees accepted the thermal environment. For the thermal sensation votes ζ_1 , a seven-point semantic differential scale from -3 to $+3$ indicating the feelings of cold, cool, slightly cool, neutral, slightly warm, warm and hot respectively was used. The results showed that no occupants voted for cold (-3), 2 for cool (-2), 15 for slightly cool (-1), 56 for neutral (0), 39 for slightly warm ($+1$), 12 for warm ($+2$) and 1 for hot ($+3$). The votes were skewed towards the warm side (i.e. 52 for $\zeta_1 > 0$ vs 17 for $\zeta_1 < 0$) with an average of 0.38. Expectedly, voters for -1 to $+1$ comprised a majority of votes. It was noted that the interviewees were willing to pay for the desired thermal comfort. Furthermore, the temperature set point and supply air flow rate of all the surveyed residential environment could be controlled by the users, a large variations on the values of V_a between 0.08 and 0.75 m s⁻¹, T_{op} between 22.8 and 31.4 °C and R_h between 67 and 100% were found as compared with air-conditioned office environment in which V_a between 0.05 and 0.31 m s⁻¹, T_{op} between 19.0 and 25.4 °C and R_h between 40 and 64% respectively [10].

Fig. 1 illustrates the measured thermal votes ζ_1 correlated with a PMV index γ [3]. Mathematical expressions of the PMV index, a function of air temperature, relative humidity, air velocity, radiant temperature, occupant metabolic rate and clothing value, i.e. $\gamma \sim \gamma(T_a, R_h, V_a, T_r, M_e, C_{cl})$, were reviewed in open literature [3,4,10]. This study, similar to some field studies of direct measurement for thermal acceptability, reported a relatively narrower operative temperature range for 80% thermal acceptability than those specified in design guidelines [4,43]. The occupant's thermal sensation in field test was sensitive as compared with the chamber test under the same environmental conditions. In additions, a 'shift' of thermal sensation vote in air-conditioned offices that occupants might prefer 'slightly cool' instead of 'neutral' was concerned. It was reported that the occupant's thermal preference in residential environment was found different from the air-conditioned environment that the occupants voted 'neutral' instead of 'slightly cool' [2,10]. It might due to the indoor environmental condition of the residential building was readjusted by the occupant to fully justify the thermal acceptance and the

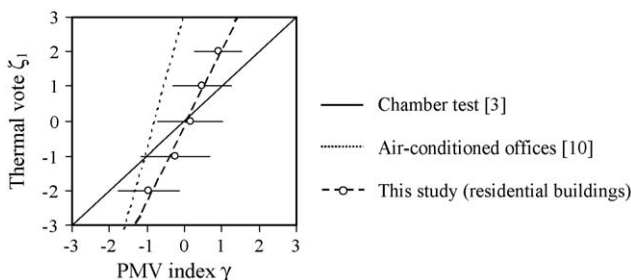


Fig. 1. Measured and predicted thermal sensation votes.

Table 2

Regression coefficients of logistic regression models.

i	Variable	$C_{0,i}$	$C_{1,i}$	$C_{2,i}$	$C_{3,i}$	$C_{4,i}$	Significance, p -value
0	ϕ_0	-33.63	22.05	1.609	11.77	21.86	0.003
1	ϕ_1	-	0.2179	0.03353	-	-	^a
2	ϕ_2	45.24	-0.0257	-	-	-	0.001
3	ϕ_3	-14.13	0.9058	-	-	-	<0.0001
4	ϕ_4	23.75	-0.2986	-	-	-	<0.0001

-: not applicable.

^a Coefficients given by Fanger [3].

energy expenditure. The correlation, with a coefficient $R = 0.987$ ($p = 0.002$, t -test), is given by,

$$\zeta_1 = 2.2\gamma - 0.15; \quad -3 \leq \zeta_1 \leq +3 \quad (5)$$

4.2. Indoor air quality, visual and aural environments

The measured ranges of CO₂ concentration (330–1500 ppm), horizontal illumination level (187–1522 lx) and equivalent sound pressure level (67–78 dBA) were approximated by geometric distributions ($p \geq 0.2$, Chi-square test). The geometric means (GMs) as well as geometric standard deviations (GSDs) for the levels of CO₂, horizontal illumination, and equivalent sound pressure respectively were: GM = 600 ppm, GSD = 1.6 ppm; GM = 112 lx, GSD = 2.9 lx; and GM = 67 dBA, GSD = 1.1 dBA. Based on the regression coefficients determined from the field measurements, the correlations between the occupant acceptance and each of the environmental parameters were studied and found to be statistically significant as presented in Table 2.

According to the survey results from 125 interviewees, 118 of them were satisfied with the IAQ, 115 with the visual environment, and 113 with the aural environment. For unsatisfactory CO₂ levels, the occupants would open a window or turn on an extraction fan for fresh air supply. The choice of illumination level was task-related and dependent on the energy cost. As occupants had little control over the noise source outside their residential buildings, they expressed significant concerns about the noise control measures. Traffic noise and noises from neighbourhood activities were the typical sources causing unsatisfactory aural environmental quality.

4.3. Indoor environmental quality

Table 3 summarizes the IEQ evaluations in terms of the acceptance parameters. The symmetric association for each parameter in the 2 by 2 ($i \times j$) table with non-zero counts n_{ij} can be expressed by Yule's Q -value, where $Q \geq 0.75$ indicates a strong relationship [44],

$$Q = \frac{n_{11}n_{22} - n_{21}n_{12}}{n_{11}n_{22} + n_{21}n_{12}} \quad (6)$$

The Q -values computed for the associations of the overall IEQ votes with the thermal environment votes, the IAQ votes, the visual environment votes and the aural environment votes were 0.97,

Table 3

Votes on acceptance of a perceiving indoor environmental quality.

IEQ	Votes	Thermal ϕ_1		IAQ ϕ_2		Visual ϕ_3		Aural ϕ_4	
		0	1	0	1	0	1	0	1
$\phi_0 = 0$	9	7	2	4	5	3	6	6	3
$\phi_0 = 1$	116	6	110	3	113	7	109	6	110
Total	125	13	112	7	118	10	115	12	113

Table 4
Overall IEQ acceptance.

Case j	Survey samples N	Survey acceptance ϕ_0	Contributors				Predicted acceptance ϕ_0
			ϕ_1	ϕ_2	ϕ_3	ϕ_4	
1	1	0	0	0	0	0	2×10^{-15}
2	0	–	0	0	0	1	8×10^{-6}
3	1	0	0	0	1	0	3×10^{-10}
4	2	0.5	0	0	1	1	0.50
5	0	–	0	1	0	0	1×10^{-14}
6	1	0	0	1	0	1	4×10^{-5}
7	2	0	0	1	1	0	2×10^{-9}
8	6	0.833	0	1	1	1	0.83
9	1	0	1	0	0	0	9×10^{-6}
10	0	–	1	0	0	1	>0.9999
11	0	–	1	0	1	0	0.55
12	2	1	1	0	1	1	1
13	0	–	1	1	0	0	5×10^{-5}
14	7	1	1	1	0	1	>0.9999
15	7	0.857	1	1	1	0	0.86
16	95	1	1	1	1	1	1

–: not surveyed.

0.94, 0.77 and 0.95 respectively, i.e. strong associations were observed.

Values of the regression constants given in Table 2 indicated the relative importance of the four contributors (i.e. thermal, IAQ, visual, aural) to the overall IEQ—larger value means greater importance. Both thermal and aural environmental qualities were deemed the most important contributors, whereas IAQ was regarded the least. Indeed, past reports on unsatisfactory residential environment in Hong Kong were mostly related to thermal comfort and noise. Apparently, occupants were very sensitive to thermal and aural comforts.

Various combinations of contributors ϕ_i for $i = 1, \dots, 4$ and the corresponding overall IEQ acceptance were considered. A total of $j = 2^4$ possibilities were found. By taking the binary notation for the acceptance (i.e. 0 for 'Unacceptable' and 1 for 'Acceptable') to rank the four contributors in the order from the most important to the least (see Table 4), the surveyed overall IEQ acceptance ϕ_0 of case j is expressed as follows, where N is the survey sample size and $N_{\phi_0=1}$ is the acceptance count,

$$\phi_{0,j} = \left(\frac{N_{\phi_0=1}}{N} \right)_j; \quad j = 1, \dots, 16 \quad (7)$$

The survey, in which merely 95 respondents accepted the indoor environment, had limited number of samples. As shown in Table 4, cases $j = 2, 5, 10, 11$ and 13 were not recorded, and only 3 cases had at least 5 samples. It was noted that cases with a small sample size would be subject to considerable uncertainty. However, good agreements were found between the predicted and observed results for all cases as illustrated in Fig. 2. The sample correlation coefficient was 0.9999 ($p < 0.0001$, t -test), and that suggested a strong linear association between the predicted and the surveyed overall IEQ acceptances.

4.4. Model sensitivity

The overall IEQ acceptance in relation to the contributors ζ_1 , ζ_2 , ζ_3 and ζ_4 was investigated via the model proposed in Eq. (1) using the regression constants listed in Table 2.

Regarding the thermal vote ζ_1 , the maximum thermal acceptance was evaluated for occupants (metabolic rate range = 1–1.2 met) residing in a thermal environment of $21^\circ\text{C} \leq T_a \leq 33^\circ\text{C}$, $20^\circ\text{C} \leq T_r \leq 39^\circ\text{C}$, $60\% \leq R_h \leq 100\%$, $0.05 \text{ m s}^{-1} \leq V_a \leq 0.85 \text{ m s}^{-1}$ corresponding to $21^\circ\text{C} \leq T_{op} \leq 36^\circ\text{C}$. As the occupants would adjust their clothing values for maximum thermal acceptance, this study

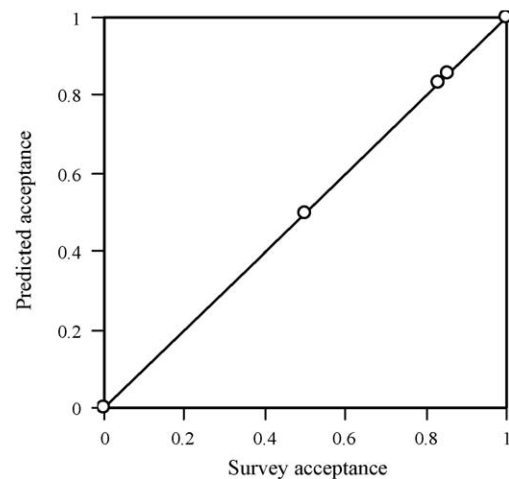


Fig. 2. Overall IEQ acceptance.

predicted their thermal responses, i.e. thermal vote ζ_1 and thermal acceptance ϕ_1 as shown in Fig. 3(b and c) respectively, which corresponded to a probable clothing value range of 0.1–1.7clo (Fig. 3(a)). According to Fig. 3, an occupant would accept the thermal environment and keep $\phi_1 > 0.9$ by adjusting his/her clothing value for an operative temperature T_{op} up to 29°C . At $T_{op} > 29^\circ\text{C}$, the thermal acceptance would decrease; and at $T_{op} > 32^\circ\text{C}$, the thermal environment would become unacceptable.

In order to examine the dependence of the predicted overall IEQ acceptance on the variations of the four contributors, $\zeta_2 = 800 \text{ ppm}$; 1800 ppm , $\zeta_3 = 15 \text{ lx}$; 50 lx , and $\zeta_4 = 50 \text{ dBA}$; 80 dBA were selected as the additional nominal conditions. Fig. 4 shows the dependency with two contributors remaining unchanged under the nominal conditions. As expected, the predicted overall IEQ acceptance was very sensitive to an operative temperature $> 28^\circ\text{C}$, but not so to the CO_2 concentration ζ_2 . The acceptance also dropped gradually as the noise level ζ_4 increased. A significant drop was found when the noise

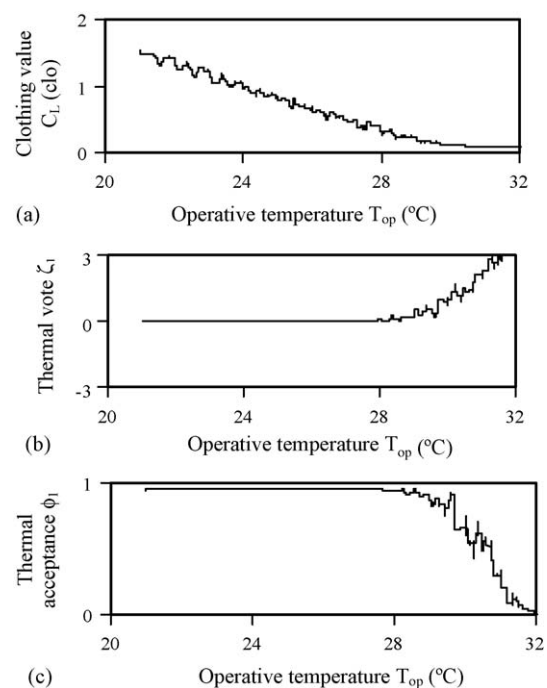


Fig. 3. Predicted occupant response for thermal environment in residential buildings: (a) clothing value at the maximum acceptance level; (b) predicted thermal vote; (c) predicted thermal acceptance.

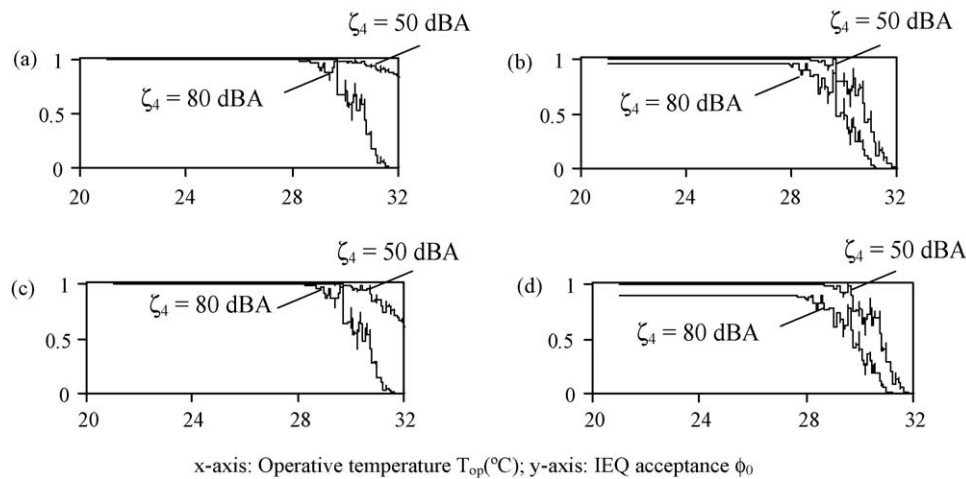


Fig. 4. Predicted occupant acceptance of IEQ in residential buildings of Hong Kong: (a) $\zeta_2 = 800$ ppm, $\zeta_3 = 50$ lx; (b) $\zeta_2 = 800$ ppm, $\zeta_3 = 15$ lx; (c) $\zeta_2 = 1800$ ppm, $\zeta_3 = 50$ lx; (d) $\zeta_2 = 1800$ ppm, $\zeta_3 = 15$ lx.

level was >70 dBA. It was reported that the overall acceptance would increase gradually starting from an illumination level ζ_3 of 10 lx and remain relatively steady at $\zeta_3 > 50$ lx.

5. Conclusion

Up until now, overall indoor environmental quality (IEQ) in terms of an occupant's acceptance has not been considered in many residential building designs. In this study, the overall IEQ of residential apartments in Hong Kong was evaluated by 125 occupants in four aspects, namely thermal comfort, indoor air quality, equivalent noise level and illumination level. The results showed that the operative temperature, CO_2 concentration, equivalent noise level and illumination level all had important effects on the overall IEQ acceptance. Based on the total votes, both thermal and aural environmental qualities were deemed the most important contributors whereas indoor air quality was considered the least. Empirical expressions were proposed to approximate the occupant acceptance. It was noted that an occupant would accept the thermal environment by adjusting his/her clothing value for an operative temperature up to 29°C . This thermal acceptance would decrease at any operative temperature higher than 29°C and the thermal environment would become unacceptable at an operative temperature above 32°C . Moreover, the predicted overall IEQ acceptance would be very sensitive to an operative temperature higher than 28°C , but not to the CO_2 concentration. A significant drop in the acceptance was also found when the noise level was exceeding 70 dBA, while the visual acceptance would increase gradually starting from an illumination level of 10 lx and remain relatively steady at 50 lx or above.

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